

Detecting Fake Reviews on Yelp Using Machine Learning

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I. Introduction

In consumer decision-making, analyzing online reviews has become an integral part of the consumer decision-making process. With platforms such as Yelp available online, millions of reviews influence where people eat, shop, and spend. However, the rise of false reviews from competitors, paid promoters or bots, continues to erode trust in these platforms.

II. Business Objectives

Fake reviews distort consumer perception and unfairly disadvantage honest businesses. For Yelp and similar platforms, unchecked review fraud undermines platform credibility and advertiser trust. This project addresses two core business needs.

1. Automatically classify Yelp reviews as genuine or fake.
2. Identify the linguistic and behavioral signals that most strongly indicate fraud.

Fraudulent reviews harm both consumers and legitimate businesses. An accurate classifier allows Yelp to proactively flag suspicious content faster than human moderation alone, protecting platform integrity and consumer trust.

Beyond a binary classifier, we aim to understand why a review is flagged. Key signals such as sentiment extremity, review length, reviewer activity level, and friend count, give Yelp actionable intelligence to refine content policies and help businesses understand what authentic engagement looks like.

Additionally, the model needs to accommodate human-centeredness to the approach. False positives can be detrimental when it comes to flagging reviews made by genuine user groups. Both functionality and transparency of the model is necessary for Yelp.

III. Results

Our best-performing model, an enhanced Random Forest trained on TF-IDF text features combined with numeric metadata (star rating, review count, friend count, review length) and a TextBlob sentiment score, achieves 84.7% accuracy and an F1 score of 0.853 on held-out test data, a significant jump from the text-only baseline (F1: 0.689, Accuracy: 66%).

Before finalizing the Random Forest, we trained a Logistic Regression model to surface which linguistic signals most distinguish genuine from fake reviews. Its coefficients revealed that genuine reviews use relational, human-centred language ("owner," "manager," "waitress"), while fake reviews over-index on menu items or hyperbolic phrases lacking authentic human context. These insights guided our feature engineering and validated that the patterns the Random Forest captured were meaningful, not just statistical noise.

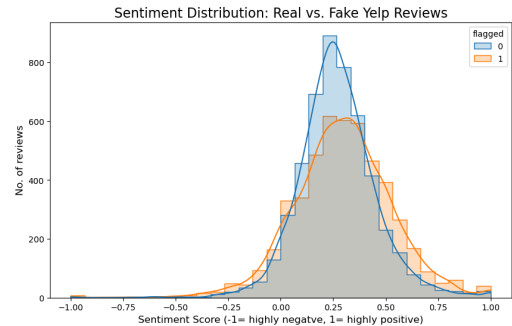


Fig. 1: Sentiment score distribution for genuine (blue, flagged=0) and fake (orange, flagged=1) Yelp reviews. Fake reviews skew toward emotional extremes, while genuine reviews cluster around neutral-to-moderate sentiment.

The final Random Forest confirms that behavioral signals are among the strongest predictors of fraud. Reviewer friendCount, reviewCount, and ReviewLength rank as top features, indicating that fake reviewers leave detectable footprints in their activity patterns beyond their writing alone. Sentiment analysis further supports this: fake reviews cluster at emotional extremes, while genuine reviews show more moderate distributions consistent with the observation that fake reviews disproportionately appear at 1-star and 5-star ratings.

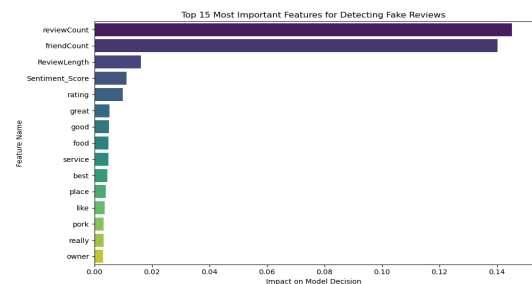


Fig. 2: Top 15 most important features identified by the enhanced Random Forest. Behavioural metadata — reviewCount, friendCount, and ReviewLength — dominate predictive power, outweighing individual text-based features.

Adding sentiment and behavioural features improved the F1 score from 0.689 to 0.853, a gain of over 23%. The enhanced Random Forest is preferred over Logistic Regression for deployment because it achieves better precision, meaning fewer authentic reviews are wrongly flagged critical, since false positives erode user trust more than occasional missed fraud.

We recommend deploying the model as a human-in-the-loop filter, flagging high-probability fake reviews for manual review rather than automatic deletion, with a conservatively tuned threshold. As a complementary measure, monitoring sentiment velocity (sudden influxes of polarized reviews) can serve as an early warning system for coordinated bot attacks or paid campaigns.